

Safety Evaluation & In-Vitro Assays for Botanical Drugs

Plan an assay so its controls, records and limitations are visible.

Lecture notes

1. Define the question and suitable model before choosing an assay.
2. Positive, negative and vehicle controls make a response interpretable.
3. Record sample preparation, concentrations, instrument settings, raw observations and calculations.
4. Discuss limitations: an in-vitro result is not proof of clinical benefit or safety in people.

Watch with purpose

While watching the YouTube lecture, pause at each workflow decision. In your notes, identify: (a) the evidence needed before acting, (b) the safety, ethics or quality control, and (c) the record that proves the step occurred.

Knowledge check

Question	Expected answer
Why include a vehicle control?	It helps distinguish the effect of the test substance from its solvent or carrier.
What should accompany a result graph?	Methods, controls, replicates, raw-data traceability and a cautious interpretation.

Practice MCQs

1. Which choice best supports reliable professional practice?
A. Relying on memory after the session B. Recording approved checks and escalating deviations C. Skipping documentation when busy D. Changing findings after review
2. What should guide an action in a supervised learning workflow?
A. A generic online claim B. The approved process, evidence and appropriate oversight C. A marketing target D. An assumption about the case

Answer key: 1-B and 2-B. The course emphasises evidence, transparent records and appropriate supervision.

Applied learning record

Use this page after watching the lecture and reading the notes. It can support a discussion with an industry mentor, faculty supervisor or placement-cell reviewer.

Reflection prompt	Your notes
Key concept explained in my own words	
Evidence or record I would create	
Safety, ethics or quality control I would check	
One question for my mentor	
How this relates to an internship or entry-level role	

Learner name: _____ Date: _____ Mentor feedback:
